SINGULAR
GENOMICS

DEERFIELD INVESTMENT COMMITTEE

The Spatial Data Engine.

An update from the operating company you backed — and a look at what is becoming possible in the next 12 months.

JOHN STARK
CHIEF EXECUTIVE OFFICER
JUNE 2026

SECTION 00

A 30-minute walkthrough — operator, market, plan, and the ask.

01 · WHO IS RUNNING THE COMPANY

John Stark

CHIEF EXECUTIVE OFFICER · SINGULAR GENOMICS

CAREER ANCHORS

affymetrix *Thermo Fisher · BD · ELITech*

Diagnostics · IVD · CDx · pharma BD · M&A

An operator with one pattern — *turn diagnostic technology into outcomes that move patient care.*

01 Drug-development centric.

Every technology gets deployed where it moves the drug forward — companion diagnostics, patient stratification, response. Pharma is how value is captured.

02 Patient impact first.

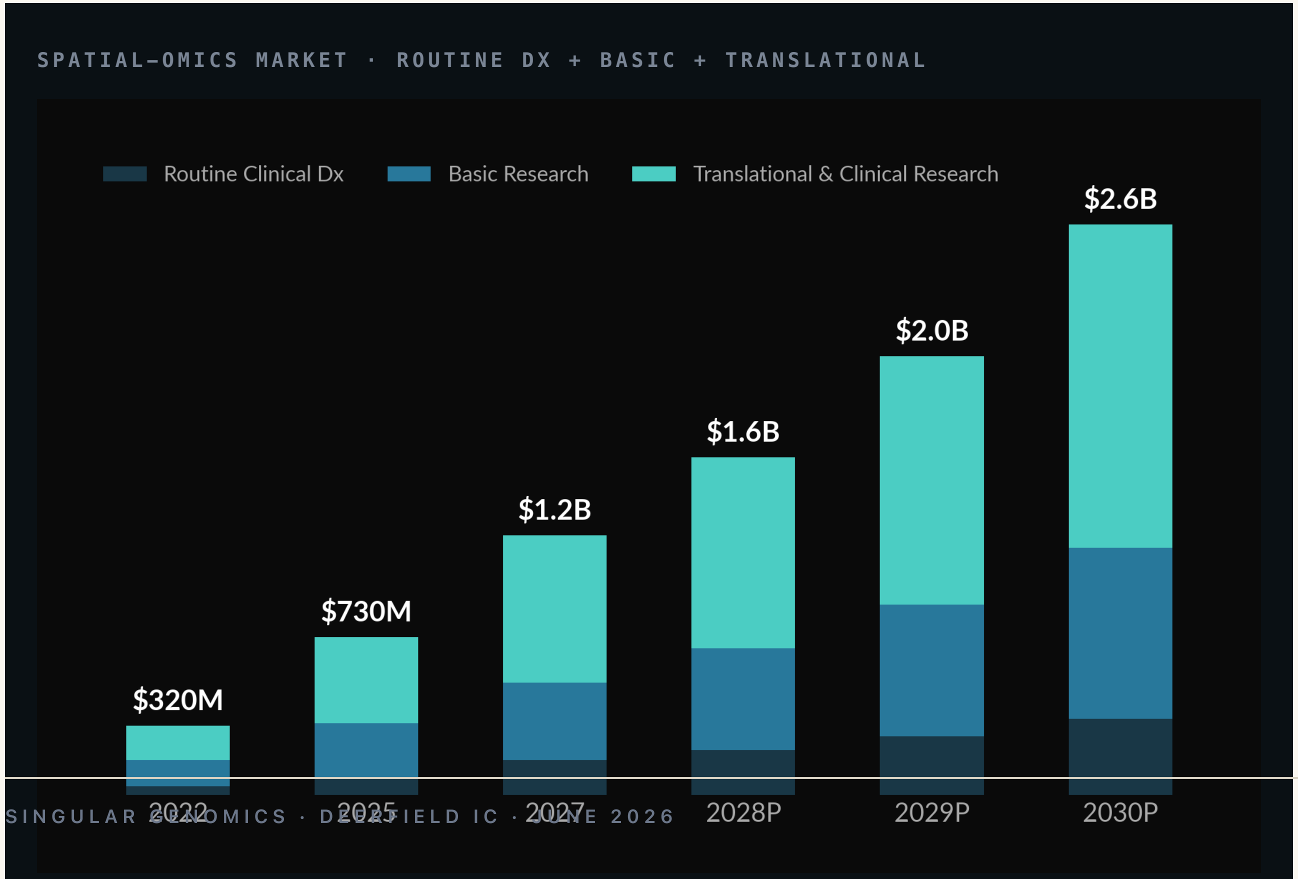
Resources flow to technology that changes what happens at the bedside. Tools without a line to a patient are deprioritized.

03 Engineering & computational mindset.

Efficiency, quantitative detection, scale. Run the math first, design the experiment second, build the org around the bottleneck the math surfaced.

The spatial market isn't a 2030 story. *It's a 2026 budget line.*

"Spatial is too early" was the 2024 instrument cycle talking. Today the kit budget is funded, cohort projects are mid-procurement, and the buyer has moved from the discovery lab to a translational program with timelines.



WHERE THE GROWTH IS

- **Translational pharma** — patient stratification and response cohorts
- **Cancer-center networks** — reflex testing, atlas builds
- **Biobank conversion** — FFPE block libraries that need to be measured
- **Companion-Dx feeders** — cohorts behind every approved targeted therapy

THE TAKE-HOME

Don't write off the spatial kits franchise. **It is the wedge into a software + pharma data business that is materially larger than the box business it grew**

Three engines, one cohort. *The hardware seeds it, the kits monetize it, the intelligence compounds it.*

G4X · THE PLATFORM

Tri-modal subcellular spatial · **128 samples/run · ~$280/sample.**

ENGINE 01

$18–29B

Instruments

Platform & consumables footprint. The cohort-seeding layer. Tools-multiples ladder: **1–3× revenue**.

ENGINE 02 · TODAY'S REVENUE

$450M

Kits & Reagents

Tri-modal consumables — recurring, per-sample, sticky. **The franchise that funds the intelligence layer.**

ENGINE 03 · THE MOAT

8–20×

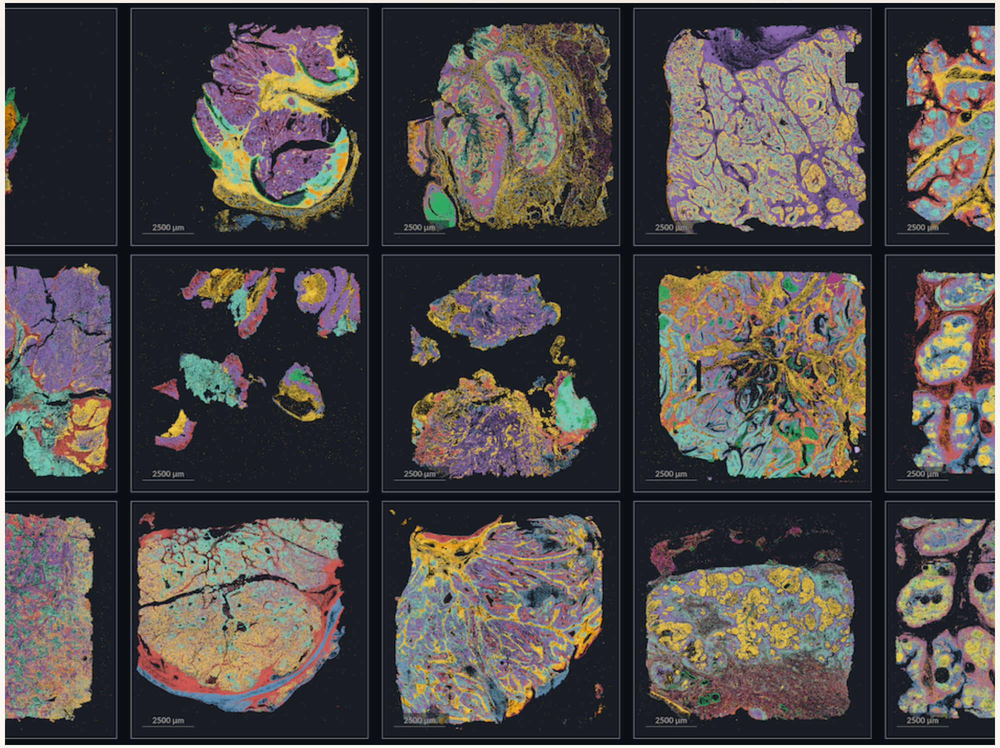
Intelligence Portfolio

Virtual IHC · QCS · API + license layer. **Data & AI multiples — where Foundation Medicine sold for $3B+.**

A funded market — and the only platform that can fulfill it at *cohort scale.*

$450M

2026 addressable spatial-kit and reagent budget across cancer centers, translational pharma, biobank conversion.



WHY WE WIN THIS MARKET

- **Tri-modal in one run.** Only commercial platform with RNA + protein + H&E on the same slide.
- **Feasibility at scale.** 128 samples / run unlocks 10K–50K-tissue atlases.
- **Every run feeds the Atlas.** Kit revenue accrues; data accrues alongside.

Five segments. *Two we lead.*

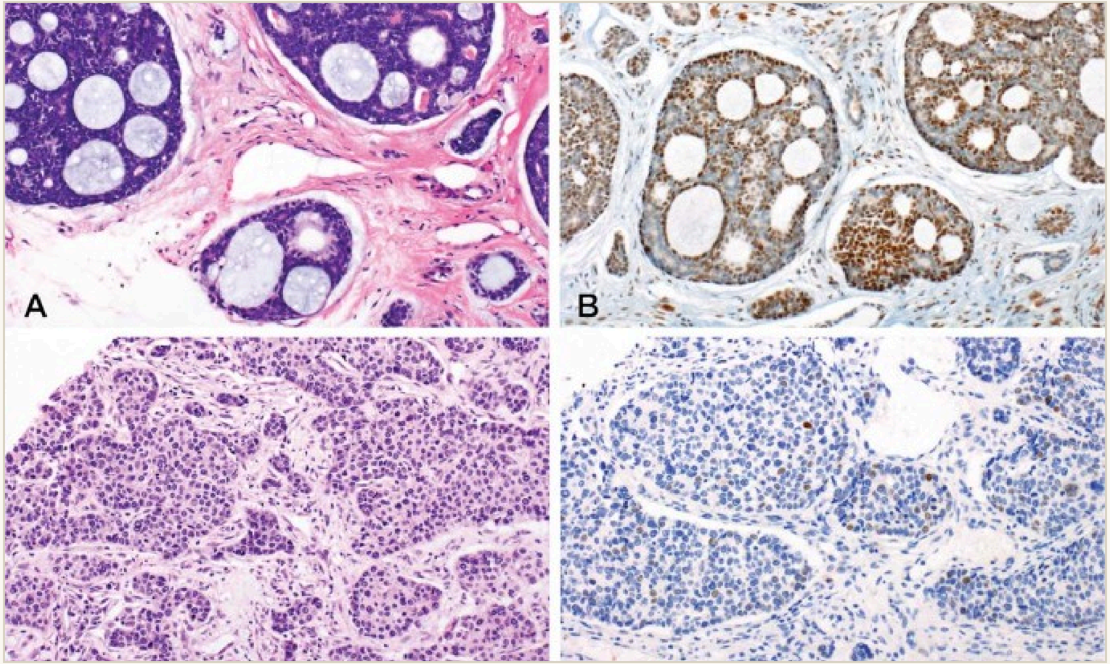
AI in medical technology is not one market — it is five. Singular sits inside the two with the highest unit economics and the strongest acquirer pull.

SEGMENT	GROWTH TO 2031	TODAY	2031
Diagnostic Imaging		$8B	$22B
WE LEAD Dx · Oncology		$12B	$38B
Surgical Devices		$6B	$18B
Digital Health		$15B	$48B
WE LEAD Drug Development		$18B	$62B

The highest TAM growth sits in the two buckets where **measured spatial ground truth is the missing input** *— and Singular is the only platform manufacturing it at cohort scale.*

TAMs from industry sources + Singular model · directional.

Diagnostics — Oncology · Drug Development. *Same data, two revenue lanes.*



LANE 01 · DX ONCOLOGY

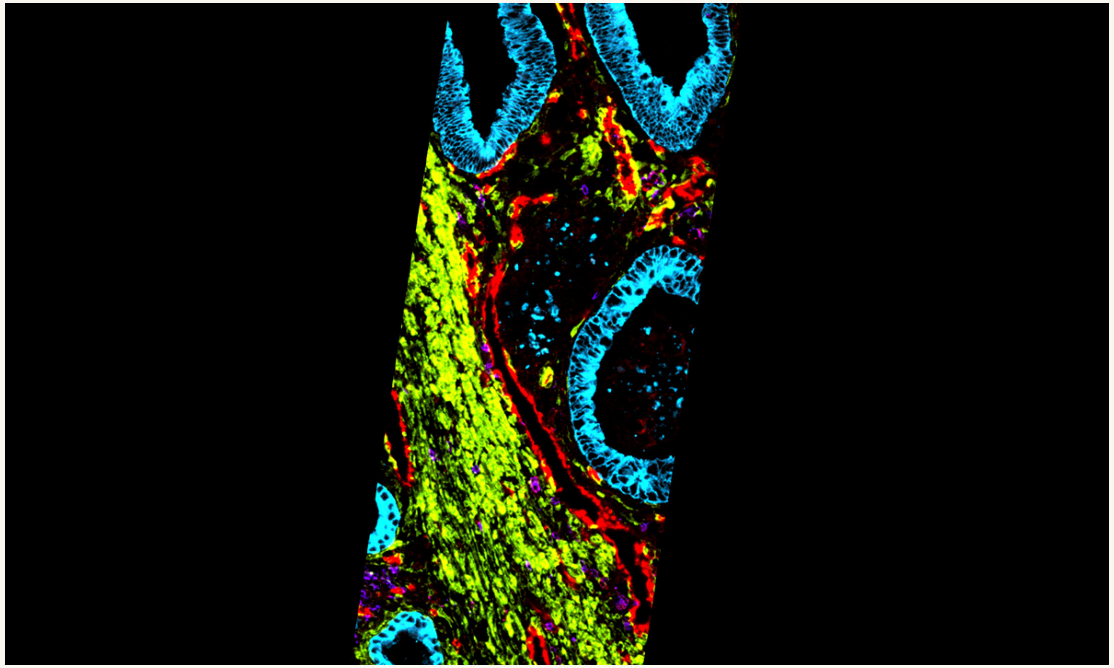
Precision pathology, at population scale.

ADDRESSABLE

~$12B today · ~$38B by 2031. Routine IHC + H&E spend at U.S./EU cancer centers.

THE PRODUCT

Virtual IHC. Predict the antibody-stain result directly from the H&E + spatial readout



LANE 02 · DRUG DEVELOPMENT

The training data pharma R&D is paying for now.

ADDRESSABLE

~$18B today · ~$62B by 2031. Trial + R&D spend on cell therapy, ADCs, bi-specifics.

THE PRODUCT

QCS. Quantitative cell scoring — proves a candidate ADC or targeted therapy is hitting at

A flywheel, not a stack. *Each layer pays for the next.*

LAYER 01 · EDGE

Platform & instruments

G4X · NGS · the cohort-seeding box.

~$420K · 128/run

→

LAYER 02 · INTEGRATION

Central Lab

Runs every sample on one protocol, one QC, one data schema · paired with clinical outcomes.

→

LAYER 03 · MARKETS

Intelligence Portfolio

Virtual IHC · QCS · model licenses · APIs & predictions.

SOFTWARE MARGINS

→

LAYER 04 · SERVICES

Customer-funded data

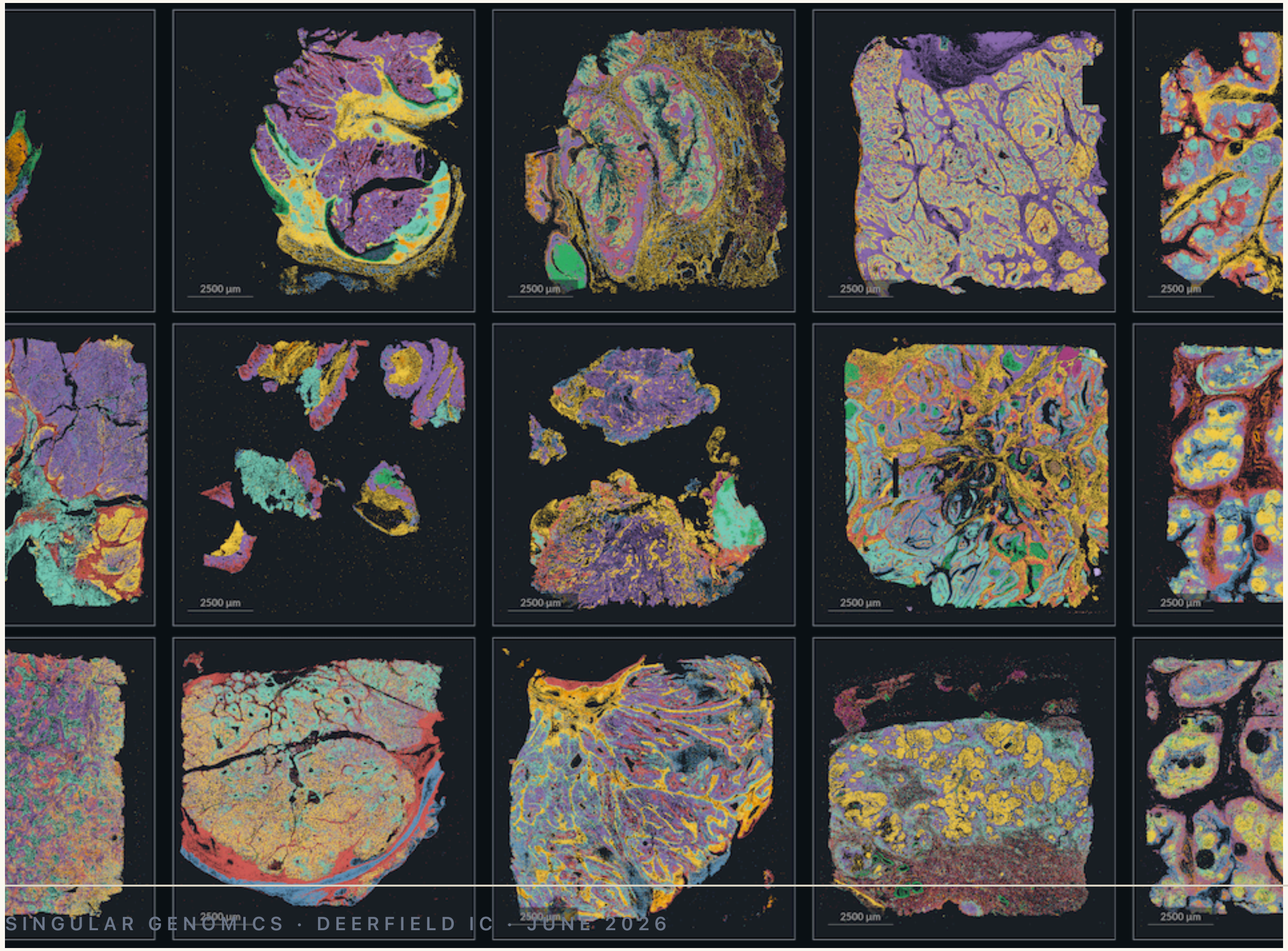
Pharma + cancer centers pay per-sample — funds the Atlas we are building anyway.

REVENUE · NEW CUSTOMERS · COHORT GROWTH

Instruments seed customers → kits and services generate cohort-scale data → the Central Lab integrates that data with clinical outcomes → Intelligence products monetize at software multiples → the cash funds more instruments and more cohorts.

The *Disease, Cell & Drug-Class Atlas.*

A single labeled library — measured RNA + protein + morphology on the same tissue, paired with the clinical outcome. Foundation Medicine sold for $3B+ with ~50K samples and sequencing alone. This is the multi-modal version.



ENTRY PHASE · FUNDED

10 – 50K

labeled tissues · breast · lung · colon · responder × pre/post-treatment

LONG HORIZON

100K+

pan-cancer · TME · immune infiltration · the asset acquirers underwrite

FOUR SAMPLE CHANNELS IN PARALLEL

- **Acquire** — block libraries (Discovery LS, Sampled)
- **Services** — pharma pays per-sample (AbbVie, Pfizer, Daiichi, BMS)
- **Translational** — float-cost partnerships (Mayo, MD Anderson, Harvard)
- **Government** — NHS, NCI, PRECISE Singapore

Eight offerings. *Three carry the next 24 months.*

Services pay the bills today and seed the Atlas; Virtual IHC productizes the predictive layer; FMI 2.0 is the platform endpoint.

NOW · 2026

Services

Pharma + center cohorts run through Central Lab. Per-sample revenue, annotated data return.

~$1–3B TAM · 35–55% GM

INTERMEDIATE · 2027

Virtual IHC

Predict IHC results from the H&E + spatial. Reflex and primary for the cancer-center workflow.

~$10–15B TAM · 70–85% GM

FUTURE · 2028+

FMI 2.0

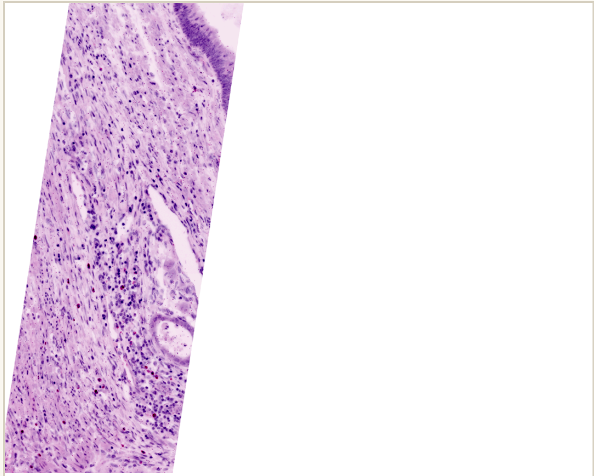
Multi-modal pan-cancer profile — the modern Foundation Medicine. Integrated into every targeted-therapy decision.

$3B+ acquirer benchmark

FULL PORTFOLIO · THE 8 OFFERINGS (HIGHLIGHTED ABOVE)

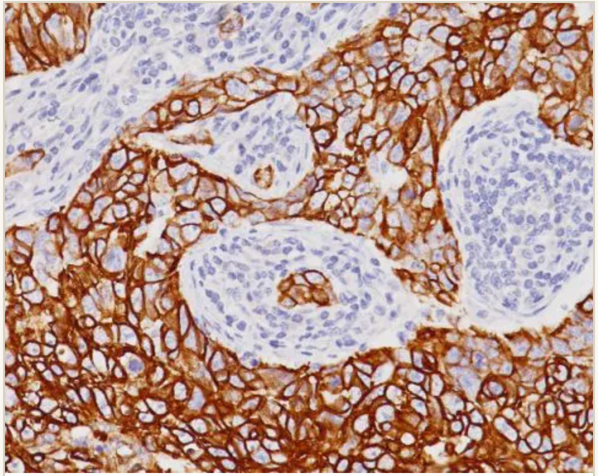
SERVICES · VIRTUAL IHC · FMI 2.0 · QCS · ADCS · SAMPLE QC · IMMUNE PROFILING · TARGET ID · VIRTUAL SCREEN

Bring *precision pathology to the masses* — not just the academic centers.



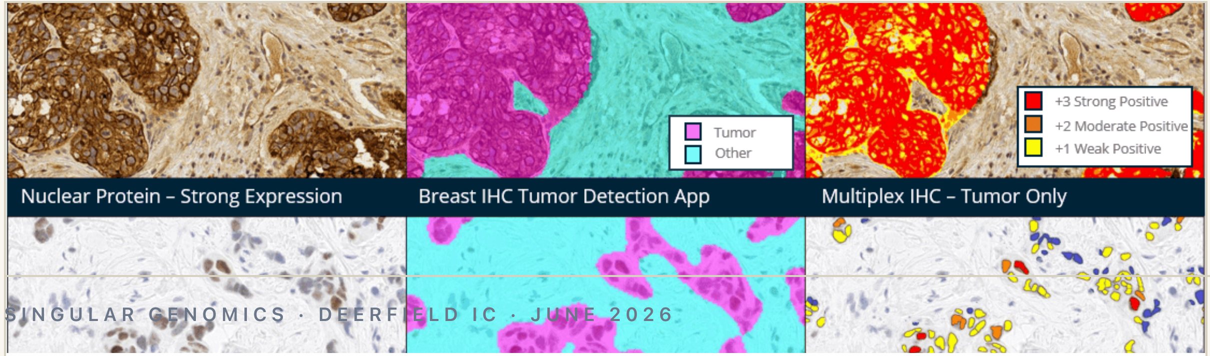
H&E · gold standard

The pink-and-purple slide every biopsy starts with. Shows what tissue *looks like*.



IHC · protein layer

An antibody stain (e.g. HER2). Slow, one-protein-at-a-time, scored by eye.



COMBINED TAM · IHC + H&E

~$8B

U.S. + EU clinical pathology spend

WHAT VIRTUAL IHC UNLOCKS

- **Cell-level detection.** Score every cell, not average a smear. Catches responders inside "non-responder" tumors.
- **No second cut.** Predict the protein-stain result from the H&E. No extra slide, no extra week.
- **Reflex + primary testing.** The intelligence becomes the first-pass call, not the confirmatory step.
- **Scales to community oncology.** The MD Anderson read, wherever an H&E is cut.

Make pharma's targeted-therapy pipelines work better. *That is where the deals are.*

MODALITY

Cell Therapy

Patient selection + persistence readout — spatial proves the cells got where they were meant to go.

MODALITY

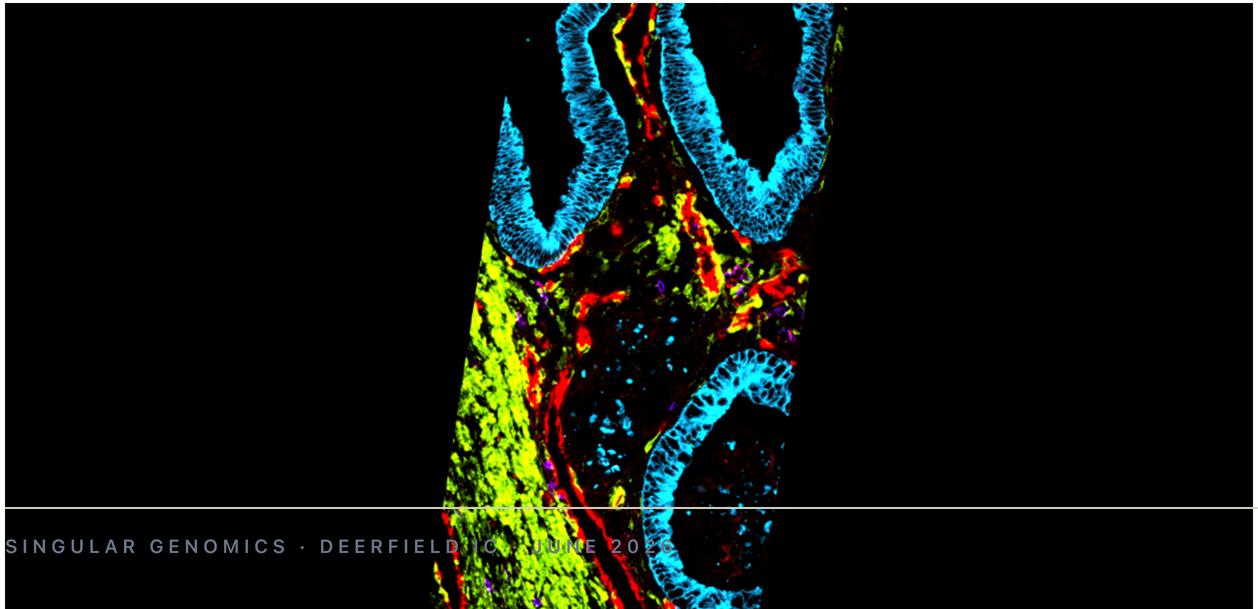
ADCs

Quantitative target expression on the tumor cell — present + how much. Drives the patient call and the dose.

MODALITY

Bi-specifics

Co-localization of two targets in the same TME — a question only spatial can answer.



WHAT WE DELIVER TO PHARMA

- **QCS.** Per-cell scoring — proves the candidate is hitting at the cell level.
- **Expanded patient reach.** Recover responders that fail conventional CDx.
- **Trial enrollment.** Screen prospective sites for the right tissue distribution before the sponsor commits.

RECENT PHARMA–AI DEALS

Roche × Recursion · *AZ × Tempus* · *Wave × Daiichi-Sankyo*

Pharma already pays for AI-driven discovery. The bottleneck is the **input layer** — measured, multi-modal, cohort-scale data on real human tissue. That is what Singular manufactures.

IN OUR PIPELINE

19/25 of top-25 pharma already in evaluation across services / kits / instruments.

From box multiples to *data & AI multiples*. Six lanes, one ownership thesis.

Tools companies trade at 1–3× revenue. Data-and-AI companies trade at 8–20×. Each acquirer family wants a different combination of the assets we are building.

COMPARABLE ACQUIRER VALUATIONS

Natera	~$30B
Tempus AI	~$10B
Isomorphic (Alphabet)	~$3–4B
Foundation Med	$3B+
Recursion + Exscientia	~$2B
PathAI (last raise)	~$350M

FMI = the structural precedent. ~50K samples + sequencing-only readout = $3B+ exit. We are building the multi-modal version of that data asset.

SIX ACQUIRER-FAMILY LANES

- **Oncology data lakes** — Tempus, Caris, FMI, Guardant
- **Digital pathology AI** — PathAI, Proscia, Indica
- **Pharma AI platforms** — Recursion, Owkin, Benevolent
- **Liquid biopsy / MRD** — Natera, Guardant, Exact Sciences
- **Big Tech in biology** — Alphabet (Isomorphic), Microsoft, OpenAI
- **Stay independent** — the FMI 2.0 endpoint

TOOLS MULTIPLES
1 – 3×

DATA & AI MULTIPLES
8 – 20×

Scenario, not forecast.

08 · SUMMARY & THE ASK

Where we are headed — and what we need from this committee.

TTM TARGET

$10M

Services + Kits

2027 FORECAST

$20M+

Adds AI products + licenses

IMPLIED SINGULAR AI VALUE

$500M+

Modeled trajectory · Q1 2027

FOCUS AREAS — NEXT 12 MONTHS

- **Pharma.** Convert the 19/25 top-pharma pipeline into 3–5 signed cohort or AI-license programs.
- **Molecular-pathology partnerships.** 3–5 cancer-center / pathology-network deals with reflex + primary intent.
- **The Atlas.** 20–50K labeled tissues across breast / lung / colon with responder and pre/post annotation.
- **Defend the $450M kit franchise.** Use it to fund the intelligence layer.

WHAT WE NEED FROM THIS COMMITTEE

- Confirmation of the path on **corporate structure & financing.**
- Alignment on **litigation timing** & counsel selection.
- Approval to move forward on **two critical hires**.

Financial targets · modeled · thought-exercise.

Appendix.

SECTION A

Detail behind every claim — operating status, competitive landscape, strategic architecture, sample-acquisition channels, the platform playbook, board close items.

Q1 2027 — the ideal press release.

A target state, not a commitment. Specific enough that the IC can sense-check the path; framed as a scenario so neither side mistakes it for guidance.

REVENUE · TTM

$10M

Services + Kits. 2027 forecast > $20M layering AI products / licenses.

PHARMA DEALS

3–5

Signed cohort, model-license, or co-development deals with top-25 pharma.

PATHOLOGY PARTNERS

3–5

Assay + analysis-interface partnerships (Harvard-class institutions).

DISEASE TISSUE ATLAS

20–50K

Labeled tissues · breast / lung / colon · responder + pre/post.

IMPLIED SINGULAR AI VALUE

> $500M

Why this matters: *puts Singular squarely on the Data & AI multiples ladder — 8–20× rather than 1–3× — and validates the strategic re-architecture the committee is being asked to fund.*

Modeled trajectory · scenario · not a commitment.

Where we sit — *9 live sites · 30+ projects · 19/25 top pharma in funnel.*

9 Live Sites

Active cohorts · multi-modal data flowing

>1,000 samples · **20+** tissue types · up to **$10K/FC** · instrument ~**$420K**

Beth Israel Lahey · Harvard Medical · Dana-Farber · Vanderbilt · Stanford · Cedars-Sinai · Einstein · U Michigan · MD Anderson · Lilly · MIT

30+ Internal Projects

In scoping / contracting / early execution

>1,000 samples · **25+** tissue types · **$15K**/flow cell · **10** custom panels

Dana-Farber · UCSD · Broad · U Michigan · Weill Cornell · Einstein · UCL · Fred Hutch · MSK · Vanderbilt · UCSF · Novartis · Gilead · Merck · Boston U · Yale · Keck USC · Cedars-Sinai · NCC Japan · MGH · Netherlands Cancer Inst.

PIPELINE · ADDED 4/30

The flywheel is filling on schedule.

INSTRUMENT OPPS	CANCER CENTERS	TOP PHARMA
~80	44 / 73	19 / 25

APPENDIX · COMPETITIVE LANDSCAPE

Spatial is a *crowded, capex-heavy, front-loaded field* — G4X is 1 of 7 commercially available platforms.

PLATFORM	MODALITY	MAX PLEX	RESOLUTION	AREA / RUN	RUN TIME	SAMPLES / YR	INSTRUMENT $	$ / RUN	INSTALL BASE
Singular G4X	RNA + protein + fH&E	500 RNA + 18 protein	Subcellular	40 cm² · 128 samples	5 days	~9,300 (theor.)	$400K + $200K HPC	~$9K · ~$280/sample	~10
10x Atera	RNA WTA + targeted	18,000 WTA + 3K	Single-cell (5 logs)	20 cm² · 4 slides	~7 days	800	<$500K	~$8.8K · $2.2K/sample	just launched
Vizgen MERSCOPE Ultra	imaging	1,000 RNA	Subcellular	1.25–3 cm² / slide	24–48 hr	~155	~$300–400K	$3.8–4.4K / slide	~107
Bruker CosMx SMI	imaging	19,000 WTX / 6K	Subcellular	3–4 cm² / slide	7–10 days	36–52	$295K	$1.85–6K	~250
Quanterix PhenoCycler-Fusion 2	protein	100+	Single-cell	~3–4 cm² whole-slide	1–2 days	~5,000	~$300–350K	$2–3K / slide	1,359 (Mar '25)
Illumina Spatial Kit (2026)	WTA via NGS	WTA	1 µm	7.5 cm²	1d + 24–48h seq	~1–2K / sequencer	$0 incremental	~$2.3–4.5K	8,400+ sequencers
Element AVITI24	NGS + cytoprofiling	WTA + protein	Single-cell	96-well plate (cells)	<24 hr	n/d	$424K new · $150K up	~$0.5–1.5K	190+

G4X retains the only triple-modality readout — RNA + protein + same-slide H&E. Throughput and per-sample economics are competitive for cohort-scale work; per-slide players are structurally locked to discovery N.

From hardware utility to data monopoly.
Three layers, one ownership thesis.

THE EDGE · DATA ENGINES

Spatial biology + sequencers
$3 – $5B
CapEx-constrained. Competitive. HW multiples.

Bulk NGS
$10 – $15B
Racing to bottom on $/Gb.

Single-cell analysis
$4 – $6B
Essential for discovery · academic funding.

Digital pathology (H&E)
$1 – $3B
Baseline hospital infra. Low margin.

TOTAL PLATFORM TAM

~$18 – 29B

Fragmented · hardware-heavy · slow growth

THE CORE · OWNERSHIP LAYER

GENERATE
Multi-modal data + clinical outcomes

INTEGRATE
× traditional clinical data

PREDICT
Foundation-model AI products

"He who owns the highest-resolution biological data, owns the future of drug development."

— FORMER MEDICAL DIRECTOR, ELI LILLY

VALUE CAPTURE · APPS

Patient monitoring & testing
$15 – $20B
QCS & Virtual IHC.

Biomarker assays + API licensing
$10 – $15B
Customized kits + API products.

In silico target discovery
$20 – $50B+
AI novel targets in spatial TME.

Virtual drug screening
$5 – $10B
Simulate efficacy on digital tissue.

TOTAL VALUE TAM

~$50 – 95B+

Software margins · embedded in pharma R&D

Four sample-acquisition channels in parallel. *Different speed, cost, rights profile.*

CHANNEL 01

Acquire

PAY FOR ACCESS

Block libraries, biobanks, retrospective oncology cohorts. Close critical gaps (rare responders, rare tumor types).

Discovery LS · Sampled

SPEED	COST
Fast	$$$
RIGHTS	**BEST FOR**
Owned	Gap-fill

CHANNEL 02

Services

CHARGE PER-SAMPLE

Pharma + cancer-center customers pay us per sample. Revenue + data dual return.

AbbVie · Pfizer · Daiichi-Sankyo · BMS

SPEED	COST
Med	Self-fund
RIGHTS	**BEST FOR**
Negotiated	Near-term rev

CHANNEL 03

Translational

FLOAT COST, SHARE DATA

Partner with leading translational institutions. We float data-gen cost; they bring samples + annotations.

Mayo · MD Anderson · UCSF · Harvard

SPEED	COST
Med	$$
RIGHTS	**BEST FOR**
Shared	TME atlas

CHANNEL 04

Government

NATIONAL-SCALE DEALS

Block archives that trade access for tech transfer + co-authored papers.

NHS · NCI · PRECISE Singapore

SPEED	COST
Slow	$
RIGHTS	**BEST FOR**
Limited	Scale

From pure tools to *platform that leads precision care + drug development* — in 12–18 months.

THE TARGET

Data & AI multiples 8× – 20×

Core IP and platform assets.

TOOLS / BOX MULTIPLES · 1× – 3× REVENUE

Foundation Medicine was acquired for **>$3B** *with ~50,000 samples. The tools-to-data trade is well-precedented.*

01 Partner to access clinical samples.

Four channels concurrently: gap-fill purchasing, 5–10 key center partnerships, services-revenue with exploratory drug-labeling, government biobanks for scale + standards.

02 Own the data layer.

Managed-cohort spatial atlas — breast, lung, colon × responder/non-responder × pre/post-treatment. Own the predictive APIs and the inputs into open-access foundation models.

03 Compete where there is more to win.

Pathology AI, oncology Dx, pharma AI, and Big Tech all need measured spatial ground truth. We supply it — or vertically integrate and become them. Either path opens kitted businesses downstream (reflex → primary).

CLOSE · BALLMER & STARK

Closing items — *review & finalize.*

- Corporate structure & financing. Confirmation of path.
- Litigation. Alignment on timing of process and selection of external firm.
- Commercial. On hold with additional HPC systems — monitoring as we progress.
- Two critical hires. Approval to move forward.

Singular Genomics · Board Meeting · May 2026

SINGULAR · GENOMICS